

LECTURE 5: MAXIMUM LIKELIHOOD AND THE INFORMATION (CRAMÉR-RAO) BOUND

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REFERENCE CB, Chapter 7.3; HMC, Chapter 6.2, or H22, Chapter 10.

In the previous lectures, we talked about evaluating estimators of θ based on data $X \sim F_X(\cdot | \theta)$, $\theta \in \Theta$, but so far we have not discussed general ways of finding them. We will now talk about one systematic way of coming up with estimators: using maximum likelihood, and we will discuss its properties. We will conclude in Section 5 by briefly discussing another method, the method of moments.

1. MAXIMUM LIKELIHOOD

Suppose the data X has density $f_X(x | \theta)$. Viewed as a function of θ , the joint probability density function (PDF) is called a likelihood function. To stress that the likelihood function views the x 's in the joint density as fixed, and is a function of θ , it is usually denoted by

$$\mathcal{L}_n(\theta | x) = f_X(x | \theta).$$

We will use the subscript n on the likelihood to emphasize this is a likelihood based on n observations. Assume that the data has density $f_X(x | \theta)$ throughout; this is without loss of generality as the following remark explains.

Remark 1. If the data is discrete, taking on values on the countable set $\mathcal{X}_D = \{x_1, x_2, \dots\}$ that doesn't depend on θ , then $\mathcal{L}_n(\theta | x) = P(X = x | \theta)$, and all results below go through. More generally, suppose the data is mixed, so that $P(X \in A | \theta) = \sum_{x \in \mathcal{X}_D \cap A} P(X = x | \theta) + \int_A g(x | \theta) dx$, with $\sum_{x \in \mathcal{X}_D} P(X = x | \theta) + \int g(x | \theta) dx = 1$. Then all the results below apply, we just have to set the “density” to be $f_X(x | \theta) = P(X = x | \theta) \mathbb{1}\{x \in \mathcal{X}_D\} + g(x | \theta) \mathbb{1}\{x \notin \mathcal{X}_D\}$. In particular, we set the likelihood to

$$\mathcal{L}_n(\theta | x) = P(X = x | \theta) \mathbb{1}\{x \in \mathcal{X}_D\} + g(x | \theta) \mathbb{1}\{x \notin \mathcal{X}_D\}.$$

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By definition, the maximum likelihood estimator (MLE) $\hat{\theta}_{ML}$ of θ maximizes the likelihood function,

$$\hat{\theta}_{ML} = \arg \max_{\theta \in \Theta} \mathcal{L}_n(\theta | x).$$

Thus, the MLE is a parameter value that gives the greatest likelihood (not probability!) of observing x . Since $\log(x)$ is strictly increasing in x , it is easy to see that $\hat{\theta}_{ML}$ also maximizes

$$\ell_n(\theta | x) := \log \mathcal{L}_n(\theta | x).$$

The function $\ell_n(\theta | x)$ is called the log-likelihood, and its derivative is called the *score*,

$$\mathcal{S}_n(\theta | x) := \frac{\partial}{\partial \theta} \ell_n(\theta | x).$$

If $\ell_n(\theta | x)$ is differentiable in θ , and $\hat{\theta}_{ML}$ lies in the interior of Θ , then $\hat{\theta}_{ML}$ satisfies $\mathcal{S}_n(\hat{\theta}_{ML} | x) = 0$.

If $X = (X_1, \dots, X_n)$ is a random sample from a distribution with density $f(x | \theta)$, then $f_X(x | \theta) = \prod_{i=1}^n f(x_i | \theta)$, and

$$\mathcal{S}_n(\theta | x_1, \dots, x_n) = \sum_{i=1}^n \frac{\partial \log f(x_i | \theta)}{\partial \theta} = \sum_{i=1}^n \frac{\partial f(x_i | \theta) / \partial \theta}{f(x_i | \theta)}.$$

Now you can see the reason why we took the log of the likelihood function: it is easier to take the derivative of the sum than the derivative of the product.

Remark 2. To save on notation, we'll typically keep the argument x implicit and just write $\mathcal{L}_n(\theta)$, $\ell_n(\theta)$ and $\mathcal{S}_n(\theta)$.

Remark 3. It is a consequence of the factorization theorem that if $\hat{\theta}_{ML}$ is unique then it is a function of *any* sufficient statistic. That is, if T is a sufficient statistic, then $\hat{\theta}_{ML}$ is a function of T (HMC, Theorem 7.3.2).¹

Digression. If the likelihood doesn't have a unique maximum, then one can always pick one that is a function of the sufficient statistic only, but there may be other estimates that maximize the likelihood which are not a function of a sufficient statistic. For example (see Romano and Siegel 1986, Example 8.13), suppose $X_i \sim \mathcal{U}[\theta - 1/2, \theta + 1/2]$. Then you can verify that $X_{(1)}, X_{(n)}$ are sufficient. But any $\hat{\theta}$ in the interval $[X_{(n)} - 1/2, X_{(1)} + 1/2]$ maximizes the likelihood. This holds for $\hat{\theta} = (X_{(1)} + X_{(n)})/2$, which is a function of a sufficient statistic only. But also for, say, $\hat{\theta} = X_{(n)} - 1/2 + (X_{(1)} - X_{(n)} + 1)/(\max\{X_{(2)}, 0\} + 1)$. $\square$

Example 1 (CB, Example 7.2.11, HMC Example 3.1.3). Let $X_1, \dots, X_n$ be a random sample from $\mathcal{N}(\mu, \sigma^2)$. Then

$$\log f(x_i | \mu, \sigma^2) = -\log \sqrt{2\pi} - (1/2) \log \sigma^2 - (x_i - \mu)^2 / (2\sigma^2).$$

So

$$\ell_n(\mu, \sigma^2) = -n \log \sqrt{2\pi} - (n/2) \log \sigma^2 - \sum_{i=1}^n (x_i - \mu)^2 / (2\sigma^2).$$

¹ We emphasize *any* since the whole data is also trivially a sufficient statistic, so it's not just that MLE is a function of *a* sufficient statistic.

The first-order conditions (FOCs) are

$$\begin{aligned}\partial \ell_n / \partial \mu &= \sum_{i=1}^n (x_i - \mu) / \sigma^2 = 0, \\ \partial \ell_n / \partial \sigma^2 &= -n / (2\sigma^2) + \sum_{i=1}^n (x_i - \mu)^2 / (2\sigma^4) = 0.\end{aligned}$$

So $\hat{\mu}_{ML} = \bar{X}_n$ and $\hat{\sigma}_{ML}^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2 / n = (n-1)S_n^2 / n$. $\square$

Example 2. Let $X_1, \dots, X_n$ be a random sample from $U[0, \theta]$. Then $f(x_i | \theta) = \mathbb{1}\{0 \leq x_i \leq \theta\} / \theta$. So

$$\mathcal{L}_n(\theta) = \frac{1}{\theta^n} \mathbb{1}\{0 \leq x_{(1)}\} \mathbb{1}\{x_{(n)} \leq \theta\}.$$

We conclude that $\hat{\theta}_{ML} = X_{(n)}$. $\square$

Example 3 (Normal censoring). Suppose we have a random sample $X_1, \dots, X_n$ such that $X_i = \max\{0, Z_i + \theta\}$, where $Z_i \sim \mathcal{N}(0, 1)$. Then by Remark 1, we set the density of X_i to $f(x | \theta) = \Phi(-\theta) \mathbb{1}\{x = 0\} + \phi(x - \theta) \mathbb{1}\{x > 0\}$, where Φ and ϕ are the standard normal cumulative distribution function (CDF) and PDF, respectively. This yields the log-likelihood $\ell_n(\theta) = \sum_{i=1}^n \mathbb{1}\{x_i = 0\} \log \Phi(-\theta) + \sum_{i=1}^n \mathbb{1}\{x_i > 0\} \log \phi(x_i - \theta)$, assuming all x_i are non-negative. $\square$

INVARIANCE A very useful property of the MLE is that it is invariant to parametrization. In particular, suppose we're interested in some function $\tau(\theta)$, and we know that the MLE of θ is $\hat{\theta}_{ML}$. Then the MLE of τ is given by $\hat{\tau}_{ML} = \tau(\hat{\theta}_{ML})$ (HMC, Theorem 6.1.2, or CB, Theorem 7.2.10).

Example 1 (continued). The MLE of the coefficient of variation $cv = \sigma / \mu$ is given by

$$\hat{cv}_{ML} = \frac{\hat{\sigma}_{ML}}{\bar{X}_n}. \quad \square$$

2. FISHER INFORMATION

For a given $\theta \in \Theta$, let $\mathcal{X}_\theta$ denote the support of X . We make the following assumption:

Assumption 1. The support $\mathcal{X}_\theta$ of X does not depend on θ , so that we can denote it by $\mathcal{X}$. Also, the log-likelihood $\ell_n(\theta | x)$ is twice continuously differentiable in θ for all $x \in \mathcal{X}$.

We can think of $\mathcal{S}_n(\theta)$ as a random variable, since it depends on the data X . Its second moment is called Fisher information,

$$\mathcal{I}_n(\theta) = E[\mathcal{S}_n(\theta) \mathcal{S}_n(\theta)'].$$

Fisher information plays an important role in maximum likelihood estimation. The theorem below gives two information equalities:

Theorem 1 (Information equalities). Under Assumption 1,

$$E[\mathcal{S}_n(\theta)] = 0,$$

and

$$(\mathcal{I}_n(\theta))_{ij} = -E \left[\frac{\partial^2 \ell_n(\theta | X)}{\partial \theta_i \partial \theta_j} \right].$$

Proof. We prove the result for $\Theta \subseteq \mathbb{R}$ to keep notation simple, though the same argument generalizes to the vector case. Since $\ell_n(\theta)$ is twice differentiable in θ , $f_X(x | \theta)$ is twice differentiable in θ as well. Differentiating (under the integral sign) the identity $\int_{\mathcal{X}} f_X(x | \theta) dx = 1$ with respect to θ yields

$$\int_{\mathcal{X}} \frac{\partial f_X(x | \theta)}{\partial \theta} dx = 0 \tag{1}$$

for all $\theta \in \Theta$. Note that

$$\mathcal{S}_n(\theta) = \frac{\partial \log f_X(x | \theta)}{\partial \theta} = \frac{1}{f_X(x | \theta)} \frac{\partial f_X(x | \theta)}{\partial \theta}. \tag{2}$$

Taking expectation of (2) and using (1) then yields

$$E[\mathcal{S}_n(\theta)] = \int_{\mathcal{X}} \frac{\partial f_X(x | \theta)}{\partial \theta} dx = 0,$$

which proves the first assertion. Differentiating (1) again yields

$$\int_{\mathcal{X}} \frac{\partial^2 f_X(x | \theta)}{\partial \theta^2} dx = 0 \tag{3}$$

for all $\theta \in \Theta$. Note that

$$\frac{\partial^2 \ell_n(\theta | x)}{\partial \theta^2} = \frac{\partial \mathcal{S}_n(\theta)}{\partial \theta} = \frac{1}{f_X(x | \theta)} \frac{\partial^2 f_X(x | \theta)}{\partial \theta^2} - \frac{1}{f_X^2(x | \theta)} \left(\frac{\partial f_X(x | \theta)}{\partial \theta} \right)^2.$$

Taking an expectation and using (3) and (2) then yields

$$\begin{aligned} E \left[\frac{\partial^2 \ell_n(\theta | X)}{\partial \theta^2} \right] &= \int_{\mathcal{X}} \frac{\partial^2 f_X(x | \theta)}{\partial \theta^2} dx - \int_{\mathcal{X}} \frac{1}{f_X(x | \theta)} \left(\frac{\partial f_X(x | \theta)}{\partial \theta} \right)^2 dx \\ &= - \int_{\mathcal{X}} \frac{1}{f_X(x | \theta)} \left(\frac{\partial f_X(x | \theta)}{\partial \theta} \right)^2 dx = -E[\mathcal{S}_n(\theta)^2] = -\mathcal{I}_n(\theta), \end{aligned}$$

which is our second result. $\square$

- If the support $\mathcal{X}$ depended on θ , then we'd need to use Leibniz rule to differentiate under the integral sign, which would mess up the result.
- Why is the second information equality true? I have no intuition, it seems like a coincidence that the curvature of the log-likelihood equals the variance of the score. As we'll see later, the result will give us a number of options for doing inference based on the MLE.

Digression. The key to Theorem 1 is differentiation under the integral sign. When are we allowed to differentiate like this? One set of sufficient conditions is as follows (see Newey and McFadden (1994, Lemma 3.6) or Durrett (2019, Theorem A.5.3)). If (i) for almost all z , the derivative $\partial a(z, \theta) / \partial \theta$ exists and is continuous in the neighborhood $\mathcal{N}$ of θ_0 , (ii) $\int \sup_{\theta \in \mathcal{N}} \|\nabla_{\theta} a(z, \theta)\| dz < \infty$, then $\int a(z, \theta) dz$ is continuously differentiable with $\nabla_{\theta} \int a(z, \theta) dz = \int \nabla_{\theta} a(z, \theta) dz$. $\square$

Remark 4 (Information for a random sample). Let us now consider Fisher information for a random sample. Let $X = (X_1, \dots, X_n)$ be a random sample from a distribution with density $f(x | \theta)$. Then the joint log-likelihood is $\ell_n(\theta | x) = \sum_{i=1}^n \log f(x_i | \theta)$. By the second information equality, it therefore follows that

$$\mathcal{I}_n(\theta) = n\mathcal{I}_1(\theta),$$

where $\mathcal{I}_n(\theta)$ is the Fisher information based on n observations. As a shorthand, we will write $\mathcal{I}(\theta)$ in place of $\mathcal{I}_1(\theta)$.

Example 4. Let us calculate Fisher information for a single sample $X \sim \mathcal{N}(\mu, \sigma^2)$, with σ^2 known. Thus, our parameter is $\theta = \mu$. The log-likelihood is $\ell_1(\mu | x) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2}(x - \mu)^2$. So

$$\frac{\partial \ell_1(\mu | x)}{\partial \mu} = \frac{x - \mu}{\sigma^2}, \quad (4)$$

and $\partial^2 \ell_1(\mu | x) / \partial \mu^2 = -1/\sigma^2$. So $-E[\partial^2 \ell_1(\mu | X) / \partial \mu^2] = 1/\sigma^2$. At the same time,

$$\mathcal{I}(\theta) = E[(\partial \ell_1(\mu | X) / \partial \mu)^2] = E[(X - \mu)^2 / \sigma^4] = 1/\sigma^2,$$

which verifies the second information equality. The first information equality is also clear by taking expectation of (4). $\square$

Example 5 (HMC, Example 6.2.1). Let us calculate Fisher information for a Bernoulli(θ) distribution. Note that a Bernoulli distribution is discrete. Following Remark 1, we use the probability mass function to derive the log-likelihood $\ell_1(\theta | X) = X \log \theta + (1 - X) \log(1 - \theta)$. So $\mathcal{S}_1(\theta) = X/\theta - (1 - X)/(1 - \theta)$ and $\partial^2 \ell_1(\theta | X) / \partial \theta^2 = -X/\theta^2 - (1 - X)/(1 - \theta)^2$. So

$$\begin{aligned} E[\mathcal{S}_1(\theta)^2] &= E[(X/\theta - (1 - X)/(1 - \theta))^2] \\ &= E[X^2/\theta^2] - 2E[X(1 - X)/(\theta(1 - \theta))] + E[(1 - X)^2/(1 - \theta)^2] \\ &= E[X/\theta^2] + E[(1 - X)/(1 - \theta)^2] = 1/\theta + 1/(1 - \theta) \\ &= \frac{1}{\theta(1 - \theta)}, \end{aligned}$$

where the third line follows since $x = x^2$, $x(1 - x) = 0$, and $(1 - x) = (1 - x)^2$ if $x \in \{0, 1\}$. At the same time,

$$\begin{aligned} -E[\partial^2 \ell_1(\theta | X) / \partial \theta^2] &= E[X/\theta^2 + (1 - X)/(1 - \theta)^2] = 1/\theta + 1/(1 - \theta) \\ &= \frac{1}{\theta(1 - \theta)}, \end{aligned}$$

as expected. $\square$

3. INFORMATION INEQUALITY

An important question in the theory of statistical estimation is whether there is a non-trivial bound such that no estimator can be more efficient, in the sense of having a lower mean squared error (MSE), than this bound. The theorem below is a result of this sort. The result is commonly known as the *Cramér-Rao bound*, but we shall follow the lead of Savage (1972) and call it the *Information inequality*, since the result is originally due to Fréchet (1943); Cramér (1946) and Rao (1945) just rediscovered it.

Theorem 2 (CB, Theorem 7.3.9, HMC, Theorem 6.2.1). Let X have a density $f_X(x | \theta)$, and suppose that Assumption 1 holds. Let $\delta(X)$ be a statistic with finite variance for all θ such that

$$\frac{\partial}{\partial \theta} E_\theta[\delta(X)] = \int_{\mathcal{X}} \delta(x) \frac{\partial}{\partial \theta} f_X(x | \theta) dx$$

(i.e., we can pass the derivative under the integral sign). Then

$$\text{var}_\theta(\delta(X)) \geq (\nabla_\theta E_\theta[\delta(X)])' \cdot \mathcal{I}_n(\theta)^{-1} \cdot \nabla_\theta E_\theta[\delta(X)].$$

where, if δ is vector-valued, the inequality is understood to hold in a positive semi-definite sense, and the (i, j) element of $\nabla_\theta E_\theta[\delta(X)]$ is given by $\partial E_\theta[\delta_j(X)] / \partial \theta_i$ (i.e., it is the transpose of the Jacobian). In particular, if $\delta(X)$ is unbiased for θ_1 , then $\text{var}_\theta(\delta(X)) \geq (\mathcal{I}_n(\theta)^{-1})_{11}$.

Proof. We prove the first result for the case with scalar-valued δ and θ . The first information equality gives $E_\theta[\mathcal{S}_n(\theta | X)] = 0$. So,

$$\text{cov}_\theta(\delta(X), \mathcal{S}_n(\theta | X)) = E_\theta[\delta(X) \mathcal{S}_n(\theta | X)] = \int_{\mathcal{X}} \delta(x) \frac{\partial f_X(x | \theta)}{\partial \theta} dx = \frac{\partial}{\partial \theta} E_\theta[\delta(X)].$$

On the other hand, by the Cauchy-Schwarz inequality,

$$\text{cov}_\theta(\delta(X), \mathcal{S}_n(\theta | X))^2 \leq \text{var}_\theta(\delta(X)) \text{var}_\theta(\mathcal{S}_n(\theta | X)) = \text{var}_\theta(\delta(X)) \mathcal{I}_n(\theta).$$

Combining the previous two displays and rearranging yields the first result.

The second result follows from the general version of the first result by noting that if δ is unbiased for θ_1 , then $\nabla_\theta E_\theta[\delta(X)] = \frac{\partial}{\partial \theta} \theta_1$. $\square$

Some consequences:

1. If we find an unbiased estimator of θ_1 with variance equal to the $(1, 1)$ element of $\mathcal{I}_n(\theta)^{-1}$, we have found the best unbiased estimator in the sense that it achieves the lowest MSE among all unbiased estimators, uniformly over $\theta \in \Theta$. Such an estimator is called uniformly minimum variance unbiased (UMVU).
2. If $\delta(X)$ is an estimator of $g(\theta)$, with bias $E[\delta(X) - g(\theta)] = b(\theta)$, and θ is scalar,

then we can write the bound as

$$\text{var}_\theta(\delta(X)) \geq \frac{(b'(\theta) + g'(\theta))^2}{\mathcal{I}_n(\theta)}.$$

Example 6. Let us calculate the bound for a random sample $X_1, \dots, X_n$ from Bernoulli(θ) distribution. We have already seen that $\mathcal{I}_1(\theta) = 1/(\theta(1-\theta))$ in this case. So Fisher information for the sample is $\mathcal{I}_n(\theta) = n/(\theta(1-\theta))$. Thus, any unbiased estimator of θ , under some regularity conditions, has variance no smaller than $\theta(1-\theta)/n$. On the other hand, let $\hat{\theta} = \bar{X}_n = \sum_{i=1}^n X_i/n$ be an estimator of θ . Then $E_\theta[\hat{\theta}] = \theta$, i.e. $\hat{\theta}$ is unbiased, and $\text{var}(\hat{\theta}) = \theta(1-\theta)/n$, which coincides with the information inequality. Thus, $\bar{X}_n$ is the UMVU estimator of θ . $\square$

Example 7 (CB, Example 7.3.13). Let us now consider a counterexample. Let $X \sim U[0, \theta]$. Then $f_1(x | \theta) = \mathbb{1}\{0 \leq x \leq \theta\}/\theta$, so that $\ell_1(\theta | x) = -\log(\theta)$ if $x \in [0, \theta]$ and $-\infty$ otherwise. Hence, ignoring boundary issues, $\mathcal{S}_1(\theta | x) = -1/\theta$, so the first information equality doesn't hold. Also, $-E[\partial \mathcal{S}_1(\theta | X)/\partial \theta] = -1/\theta^2 < 0$, while the Fisher information is positive, $E[\mathcal{S}_1^2(\theta | X)] = 1/\theta^2$. Thus, the second information equality does not hold either. The reason is that the support of the distribution depends on θ . Moreover, consider an estimator $\hat{\theta} = ((n+1)/n)X_{(n)}$ of θ based on a random sample $X_1, \dots, X_n$. Then $E_\theta[\hat{\theta}] = \theta$ and one can show that

$$\text{var}(\hat{\theta}) = ((n+1)^2/n^2) \text{var}(X_{(n)}) = \theta^2/(n(n+2)).$$

So $\hat{\theta}$ is unbiased, but its variance is smaller than $1/\mathcal{I}_n(\theta) = \theta^2/n$. Thus, the information bound does not hold in this example either, because it requires that the support $\mathcal{X}$ is independent of the parameter.

This example also shows that if the support of X does depend on θ , this actually makes the problem easier: while the information equality and inequality do not apply, the “information” about θ accumulates at a faster rate than usual in the sense that the convergence rate of $\hat{\theta}$ is, by Chebyshev's inequality, equal to n , not $\sqrt{n}$. $\square$

As the next section shows, the information bound is possible to attain only in a limited number of cases. So why is the information inequality useful? We will show in the next lecture that the MLE achieves the bound in large samples.

4. ATTAINABILITY OF THE BOUND (OPTIONAL)

Can we always attain the information bound in the sense that there exists an unbiased estimator of θ_1 with variance equal to $(\mathcal{I}_n(\theta)^{-1})_{11}$?

Lemma 3. Suppose conditions of Theorem 2 hold. Suppose $\delta(X)$ is unbiased for θ_1 . Then the

information inequality holds with equality if and only if $\delta(X)$ can be written

$$\delta(X) = \theta_1 + (\mathcal{I}_n(\theta)^{-1} \mathcal{S}_n(\theta | X))_1. \quad (5)$$

Proof. We give the proof for the case with scalar-valued θ . To attain the information bound, the crucial step in the derivation of the bound was the Cauchy-Schwarz inequality $\text{cov}_\theta(\delta(X), \mathcal{S}_n(\theta | X))^2 \leq \text{var}(\delta(X)) \mathcal{I}_n(\theta)$. Thus, an unbiased estimator will attain the information inequality if and only if the Cauchy-Schwarz inequality holds as an equality, which happens if and only if $\delta(X)$ and $\mathcal{S}_n(\theta | X)$ are linearly dependent. In this case there exist functions $a(\theta)$ and $b(\theta)$ such that $\mathcal{S}_n(\theta | X) = a(\theta)(\delta(X) - b(\theta))$. By the first information equality, $E[\mathcal{S}_n(\theta | X)] = 0$. Since $\delta(X)$ is unbiased, $b(\theta) = \theta$. Furthermore, by the first information equality, $\mathcal{I}_n(\theta) = \text{var}(\mathcal{S}_n(\theta | X)) = a(\theta)^2 \text{var}(\delta(X)) = a(\theta)^2 / \mathcal{I}_n(\theta)$, so that $a(\theta) = \mathcal{I}_n(\theta)$. Therefore, the estimator has to take the form $\delta(X) = \mathcal{S}_n(\theta | X) / \mathcal{I}_n(\theta) + \theta$. $\square$

Thus, to check whether the information bound can be attained, we compute the right-hand side (RHS) of eq. (5). If it doesn't depend on θ , we know it can be attained, and we also obtain the estimator that attains it at the same time! While this sounds great, it turns out that attainment is only possible in a limited range of cases: only if the underlying family of distributions constitutes what's called an exponential family (see Chapter 2 in Lehmann and Casella (1998) for references).

Example 8 (CB, Example 7.3.14, 7.3.16). As an example, let $X_1, \dots, X_n$ be a random sample from a $\mathcal{N}(\mu, \sigma^2)$ distribution, and put $\theta = (\mu, \sigma^2)'$. Then the log-likelihood equals

$$\ell_n(\theta | X) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2,$$

which implies

$$\frac{\partial \ell_n(\theta | X)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\sum_{i=1}^n (X_i - \mu)^2}{2\sigma^4} = \frac{n}{2\sigma^4} \left(\sum_{i=1}^n (X_i - \mu)^2 / n - \sigma^2 \right),$$

and

$$\frac{\partial \ell_n(\theta | X)}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu) = \frac{n}{\sigma^2} (\bar{X}_n - \mu).$$

Also,

$$\mathcal{I}_n(\mu, \sigma^2) = \begin{pmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{pmatrix}.$$

For estimating μ , the RHS of eq. (5) yields

$$\mu + \frac{\sigma^2}{n} \frac{\partial \ell_n(\theta | X)}{\partial \mu} = \bar{X}_n.$$

Therefore, the sample mean attains the information bound in general, so $\bar{X}_n$ is the best unbiased estimator of μ .

What about σ^2 ? Then, regardless of whether μ is known, the RHS of eq. (5) yields

$$\sigma^2 + \frac{2\sigma^4}{n} \frac{\partial \ell_n(\theta \mid X)}{\partial \sigma^2} = \sigma^2 + \sum_{i=1}^n (X_i - \mu)^2 / n - \sigma^2 = \sum_{i=1}^n (X_i - \mu)^2 / n.$$

This doesn't depend on σ^2 , so that the estimator $\hat{\sigma}^2(X) = \sum_{i=1}^n (X_i - \mu)^2 / n$ is UMVU for σ^2 . However, the expression does depend on μ , so that when μ is not known, it is not possible to construct an unbiased estimator of σ^2 that attains the bound. $\square$

5. METHOD OF MOMENTS

This is another general method for constructing estimators. Let $X_1, \dots, X_n$ be a random sample from some distribution $F_X(\cdot \mid \theta)$. The dimension of θ may be large, even infinite-dimensional. Write $\theta = (\beta, \eta)$, where β is a parameter of interest with dimension k and η is a nuisance parameter.

Suppose we are able to come up with a system of equations

$$E[g(X_i, \beta)] = 0,$$

where $g: \mathcal{X} \times \mathbb{R}^k \rightarrow \mathbb{R}^k$. Then the method of moments estimator $\hat{\beta}_{MM}$ of β is the solution of the above system of equations if we replace the population expectation by the sample average

$$\frac{1}{n} \sum_i g(X_i, \hat{\beta}_{MM}) = 0.$$

We assume here that the solution exists and is unique.

In the simplest case, the moments have the separable form $g_j(X_i, \beta) = X_i^j - f_j(\beta)$.

Example 9 (CB, Example 7.2.1). Let $X_1, \dots, X_n$ be a random sample from $\mathcal{N}(\mu, \sigma^2)$. Then $E[X_i] = \mu$ and $E[X_i^2] = \mu^2 + \sigma^2$. Thus, $\hat{\mu}_{MM} = \sum_{i=1}^n X_i / n$ and $\hat{\mu}_{MM}^2 + \hat{\sigma}_{MM}^2 = \sum_{i=1}^n X_i^2 / n$. So $\hat{\sigma}_{MM}^2 = \sum_{i=1}^n X_i^2 / n - (\sum_{i=1}^n X_i / n)^2$. These estimates are the same as maximum likelihood estimates of μ and σ^2 . So why use method of moments? Suppose we drop the normality assumption, and assume only that X_i are i.i.d., with mean μ and variance σ^2 , so that our model is $X_i = \mu + \sigma Z_i$, $Z_i \sim F$, and F can be any distribution with zero mean and unit variance. Then we can't use maximum likelihood to estimate μ and σ^2 , but the method of moments still yields identical estimates as before. $\square$

Example 10 (CB, Example 7.2.2). Let $X_1, \dots, X_n$ be a random sample from $\text{Binomial}(k, p)$, with both k, p unknown. Now,

$$P(X_i = j) = \binom{k}{j} p^j (1-p)^{k-j}.$$

So $E[X_i - kp] = 0$ and $E[X_i^2] = kp(1-p) + k^2 p^2$. The first equation implies $\hat{p}_{MM} \hat{k}_{MM} =$

$\bar{X}_n$. Plugging this into the second equation and letting $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}_n^2$, we get

$$\frac{1}{n} \sum_{i=1}^n X_i^2 = \bar{X}_n(1 - \hat{p}_{MM}) + \bar{X}_n^2 \implies \hat{p}_{MM} = 1 - \frac{\hat{\sigma}^2}{\bar{X}_n},$$

and thus $\hat{k}_{MM} = \frac{\bar{X}_n^2}{\bar{X}_n - \hat{\sigma}^2}$. Note that it's possible to get negative estimates of k and p , and that in general, $\hat{k}_{MM}$ will not be an integer.

Now consider the MLE. The likelihood is

$$\ell_n(k, p) = \sum_i \log \binom{k}{X_i} + n\bar{X}_n \log p + n(k - \bar{X}_n) \log(1 - p).$$

Observe that for any choice of k , the likelihood is maximized over p at $\hat{p}(k) = \bar{X}_n/k$. Plugging this back into the likelihood and maximizing over k yields

$$\hat{k}_{ML} = \operatorname{argmax}_{k=X_{(n)}, X_{(n)}+1, \dots} \ell_n(k, \bar{X}_n/k).$$

We know that $\hat{k}_{ML}$ will equal at least $X_{(n)}$ (otherwise, the log-likelihood evaluates to $-\infty$), but there is no closed form solution. Substituting back into the expression for p gives $\hat{p}_{ML} = \bar{X}_n/\hat{k}_{ML}$. It is always the case that $\hat{p}_{ML} \in [0, 1]$ and $\hat{k}_{ML}$ is a non-negative integer. $\square$

Remark 5. The previous example illustrates a general fact that the MLE is always inside Θ (Why?). By contrast, the method of moment estimates may lie outside Θ .

The idea of the method of moments is old, dating back to at least Pearson in the 19th century. There is a generalization of it which allows for more moments than the dimensionality of the parameter. It is called generalized method of moments (GMM) and will be studied extensively later on, as the main workhorse of econometrics.

Remark 6. An immediate consequence of Theorem 1 is that the maximum likelihood estimator based on a random sample $X_1, \dots, X_n$ can be viewed as a method of moments estimator based on the equality $E[S_1(\theta | X_i)] = 0$.

Remark 7. You have likely seen many method of moments estimators before, even if you haven't realized it. For example, the ordinary least squares estimator is a method of moments estimator based on the moment condition $E[X_i(Y_i - X_i'\beta)] = 0$. The instrumental variables estimator is also a method of moments estimator.

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